

Minimum stabilizer weight of periodic products of cycles

Working proof for independent audit

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Status. This draft gives an analytic mixed-side classification, using the cited Hamming-graph code theorem for one base family. For dimension at least three, one side of length at least five suffices for full distance and singleton-only minima. All new finite checks are independent corroboration rather than proof dependencies. Publication novelty has not been established; the equal-side two-dimensional distance-five family is already in Ref. [4].

Let $G = \square_{i=1}^D C_{L_i}$, with vertex set $V = \prod_i \mathbb{Z}/L_i\mathbb{Z}$, and adjacency matrix A over $\mathbb{F}_2$. The canonical graph-state generators are $K_v = X_v \prod_{u \sim v} Z_u$. For a nonempty generator subset S , the product $K_S = \prod_{v \in S} K_v$ has binary Pauli coordinates (x, Ax) , where $x = 1_S$. Consequently

$$W(S) := \text{wt}(K_S) = |\text{supp}(x) \cup \text{supp}(Ax)| = |S \cup \text{Odd}(S)| \geq |S|. \quad (1)$$

Here $\text{Odd}(S)$ contains vertices with an odd number of neighbors in S ; vertices of S remain in the support even when their Z coordinates cancel.

Theorem 1. *If $D \geq 1$ and every $L_i \geq 5$, then $\min_{\emptyset \neq S \subseteq V} W(S) = 2D + 1$. The graph state is exactly $2D$ -uniform. The bound applies to unequal side lengths.*

Lemma 2 (Two omitted consecutive slices). *Suppose that, in each coordinate, the projection of a nonempty S omits two consecutive cycle positions. Then $W(S) \geq |S| + 2D$.*

Proof. Relabel each coordinate so that positions $L_i - 2, L_i - 1$ are omitted. Thus S lies in the integer box $\prod_i [0, L_i - 3]$. For each i choose $v_i^+, v_i^- \in S$ whose i th coordinates are respectively maximal and minimal, and set $q_i^+ = v_i^+ + e_i$, $q_i^- = v_i^- - e_i$, interpreted cyclically. Each $q_i^\pm$ is outside S and has exactly one neighbor in S , namely $v_i^\pm$. Indeed, all neighbors obtained in another coordinate retain an i th coordinate outside the projection of S . The second neighbor in coordinate i is beyond the corresponding extremum, or, on wrapping around the cycle, in one of the two omitted slices.

These $2D$ witnesses are distinct. The positive and negative witnesses in the same coordinate have different i th coordinates: if the extrema are $a \geq b$, their coincidence would require $a - b + 2$ to be a multiple of L_i , whereas $2 \leq a - b + 2 \leq L_i - 1$. For $i \neq j$, witness $q_i^\pm$ has its i th coordinate outside the projection of S , while $q_j^\pm$ inherits that coordinate from a vertex of S . Therefore all witnesses lie in $\text{Odd}(S) \setminus S$, proving the claim. This argument does not require S to be connected. $\square$

Lemma 3 (Slice inequality). *Write $G = H \square C_L$, and suppose every nonzero $y \in \mathbb{F}_2^{V(H)}$ satisfies $|\text{supp } y \cup \text{supp}(A_H y)| \geq d_H$. For $x = 1_S$, write x_t for slice t , $r_t = |\text{supp } x_t|$, $P = \{t : r_t > 0\}$, and $k = |P|$. Then*

$$W(S) \geq k d_H - \sum_{u \in P} \deg_{C_L[P]}(u) r_u. \quad (2)$$

Proof. The Z coordinate in slice t is $A_H x_t + x_{t-1} + x_{t+1}$. For any binary vectors a, b, c , perturbing b to $b + c$ can remove at most $|\text{supp } c|$ elements from $\text{supp } a \cup \text{supp } b$. For occupied t , therefore,

$$w_t := |\text{supp } x_t \cup \text{supp}(A_H x_t + x_{t-1} + x_{t+1})| \geq d_H - r_{t-1} - r_{t+1}.$$

Sum over occupied t and discard the nonnegative contributions of empty slices. Reindexing the subtracted terms gives (2). Cancellations may overlap or be counted more than once; this only weakens this lower bound and does not presume disjoint cancellation sites. $\square$

Proof of Theorem 1. A singleton S has weight $2D + 1$, so it suffices to rule out $W(S) \leq 2D$. Induct on D .

For $D = 1$, a putative $W(S) \leq 2$ implies $1 \leq |S| \leq 2$. On C_L with $L \geq 5$ there are at least three empty positions. If no two empty positions were consecutive, the successor of every empty position would be occupied; the successor map is injective, so there would be at least as many occupied as empty positions. This contradicts $|S| \leq 2$. Lemma 2 now gives $W(S) \geq |S| + 2 \geq 3$.

For $D \geq 2$, assume the theorem in dimension $D - 1$ and let $r = |S| \leq W(S) \leq 2D$. Fix any coordinate. Its slice graph H satisfies $d_H = 2D - 1$ by induction. Let P and k be as in Lemma 3.

If $k \geq 5$, every induced degree is at most two, whence

$$W(S) \geq k(2D - 1) - 2r \geq 5(2D - 1) - 4D = 6D - 5 > 2D.$$

If $k = 4$, then P is a proper subset of a cycle of length at least five. Its induced graph is a nonempty disjoint union of paths, including possible isolated vertices. Hence $\sum_{u \in P} (2 - \deg_{C_L[P]} u) \geq 2$. Since each $r_u \geq 1$, its weighted counterpart is also at least two. Thus

$$W(S) \geq 4(2D - 1) - (2r - 2) \geq 4D - 2 > 2D.$$

Therefore every coordinate projection has at most three occupied positions.

Suppose a coordinate projection has no two consecutive empty positions. Again the successor injection gives $L - k \leq k$. Since $L \geq 5$ and $k \leq 3$, this forces $k = 3$ and $L \in \{5, 6\}$. For $L = 5$ the two empty positions are nonadjacent, and the occupied induced graph consists of one edge and one isolated vertex. For $L = 6$, occupied and empty positions alternate, so every occupied induced degree is zero. In either case the induced maximum degree is at most one. The slice inequality yields

$$W(S) \geq 3(2D - 1) - r \geq 4D - 3 > 2D.$$

All inequalities are strict also in the smallest induction dimension $D = 2$: the three respective lower bounds are 7, 6, 5, each exceeding 4.

Consequently every coordinate has two consecutive empty slices, and Lemma 2 gives $W(S) \geq r + 2D \geq 2D + 1$, a contradiction. Finally, a graph-state reduced density matrix is maximally mixed on every set of at most t qubits exactly when no nonidentity stabilizer has weight at most t , giving the uniformity statement. $\square$

Corollary 4 (Minimum-weight multiplicity). *For $D \geq 3$ and all $L_i \geq 5$, the only stabilizer elements of weight $2D + 1$ are the $N = \prod_i L_i$ individual canonical generators. In particular the minimum-weight stabilizer multiplicity is exactly N .*

Proof. Use the established theorem for the slice graphs and assume $W(S) \leq 2D + 1$, so $r \leq 2D + 1$. The preceding three projection bounds become respectively

$$\begin{aligned} k \geq 5 : \quad & W(S) \geq 6D - 7, \\ k = 4 : \quad & W(S) \geq 4D - 4, \\ k \leq 3 \text{ with no adjacent empties} : \quad & W(S) \geq 4D - 4. \end{aligned}$$

For $D \geq 3$ all exceed $2D + 1$ (at $D = 3$ they are 11, 8, 8, compared with 7). Thus Lemma 2 applies and $r + 2D \leq W(S) \leq 2D + 1$, forcing $r = 1$. Distinct generator subsets specify distinct stabilizer elements, so precisely N elements attain this weight. $\square$

Why the dimension restriction matters. On $C_5 \square C_5$, let $S = \{(t, t) : t \in \mathbb{Z}/5\mathbb{Z}\}$. Every vertex has either zero or two neighbors in S , so $\text{Odd}(S) = \emptyset$ and $W(S) = 5$, although $|S| = 5$. The two diagonal slopes and their five translates supply ten such stabilizers. The exhaustive calculation below confirms that these and the 25 individual generators are all the weight-five elements.

Corollary 5 (Sharp universal common-side threshold). *The number five cannot be replaced by four in Theorem 1.*

Proof. On $C_4^{\square D}$ write $A = \sum_i (T_i + T_i^{-1})$ for the coordinate shifts over $\mathbb{F}_2$. These commute and $T_i^4 = I$, so

$$A^2 = \sum_i (T_i^2 + T_i^{-2}) = 0.$$

For $x = Ae_0$, the support consists of the $2D$ distinct neighbors of the origin, while $Ax = 0$. Thus $W(\text{supp } x) = 2D$. This proves sharpness of a universal lower threshold for all side lengths; it does not classify tori with some side lengths four and others larger. $\square$

Reproducible finite checks. The accompanying C++ program computes $|x \vee Ax|$ by exact bit operations for every generator subset of each stated cardinality containing the origin. Translation transitivity implies that a minimum of any fixed cardinality occurs among these anchored subsets. Every subset of weight at most b has cardinality at most b by (1); there is no connectivity restriction. The number of enumerated anchored subsets must equal $\binom{N-1}{r-1}$. The raw log and structured JSON retain these counts, witnesses, minimizer counts, timings, compiler identity, and commands. For a vertex-transitive graph, an anchored minimizer count a_r corresponds to Na_r/r global minimizers, by counting incidences between minimizing subsets and their vertices.

For C_5^3 all 4,700,324,626 anchored subsets of cardinalities one through seven were checked; the minimum weights by cardinality are 7, 10, 13, 14, 15, 12, 13. This exhaustively verifies both distance seven and the 125-element minimum multiplicity in this instance. For C_5^2 , cardinalities one through five have minima 5, 6, 7, 8, 5, with one and two anchored minimizers in cardinalities one and five, respectively, giving 35 global weight-five elements. Checks for $C_5 \square C_6$ and $C_5 \square C_7$ exhaust cardinalities up to four. The C_5^4 check exhausts only cardinalities up to four and therefore is a partial stress test, not a finite certification of its distance nine.

Literature position and unresolved priority. Sudevan et al. [1], Sec. II.B and Appendix C, prove the periodic cluster-state result for linear size at least eight and explicitly conjecture the threshold five from low-dimensional evidence. The October 2025 version of the subsequent code paper [2], Proposition 2 and Corollary 1, retains the side-length-at-least-eight hypothesis. The proof above, if independently accepted, resolves the cited arbitrary-dimensional threshold-five conjecture. The two-dimensional rectangular distance-five case and a diagonal minimum on the five-by-five torus already appear in Ref. [8]. This is a mathematical comparison, not a confirmed novelty claim. In particular the 2019 abstract [3] explicitly mentions studying Cartesian products of diagonal distance without stating a usable theorem. The associated full result was not located in the bounded search; it remains a priority-check lead.

Mixed side lengths: stronger classification

The preceding proof assumes every side is long. The extension below allows short factors. All new arguments below are analytic. The finite arithmetic certificates and large graph scans are independent corroboration rather than necessary proof steps. Write $d(H) = \min_{x \neq 0} |\text{supp } x \cup \text{supp}(A_H x)|$.

Theorem 6 (Mixed-side classification). *Let $D \geq 3$ and $L_i \geq 3$ for every coordinate. Then*

$$d\left(\bigoplus_{i=1}^D C_{L_i}\right) = \begin{cases} 2D, & \text{if every } L_i \in \{3, 4\}, \\ 2D + 1, & \text{if at least one } L_i \geq 5. \end{cases}$$

In the second case, the only minimum-weight stabilizers are the $N = \prod_i L_i$ canonical generators. Thus the state is exactly $2D$ -uniform if and only if at least one side is long.

Two product lemmas

Suppose H is h -regular and $d(H) \geq h$. In $H \square C_L$, a singleton slice has internal weight $h + 1$. With $r_t = |\text{supp } x_t|$, occupied slices obey

$$W_t \geq \max\{r_t, h + 1_{r_t=1} - r_{t-1} - r_{t+1}\}, \quad (3)$$

while empty slices have $W_t = |\text{supp}(x_{t-1} + x_{t+1})| \geq |r_{t-1} - r_{t+1}|$. Put $r = \sum_t r_t$, let k be the number of occupied slices, and define $\delta = \sum_{t \in P} (2 - \deg_P t) r_t$. Then

$$W \geq kh - 2r + \delta + \#\{t : r_t = 1\} + \sum_{t \notin P} W_t. \quad (4)$$

If exactly two slices u, v are occupied, then

$$W \geq w_H(u) + w_H(v) \geq 2h. \quad (5)$$

For nonadjacent slices their internal outputs are unchanged. For adjacent slices, their perturbations can remove at most $|\text{supp } v \setminus \text{supp } u| + |\text{supp } u \setminus \text{supp } v| = |\text{supp}(u + v)|$ support sites in total. For $L = 3$ the remaining slice restores exactly that many sites. For $L \geq 4$ the two outward empty slices restore $|u| + |v|$, at least this amount. This argument also proves the first inequality without the hypothesis $d(H) \geq h$.

Lemma 7 (Four-cycle lifting). *If $h \geq 2$ is even, H is h -regular, and $d(H) \geq h$, then $d(H \square C_4) \geq h + 2$.*

Proof. Assume $W \leq h + 1$, hence $r \leq h + 1$. With one occupied slice, $W = w_H(x) + 2r \geq h + 2$. With two, (5) gives $2h > h + 1$. With four, $h = 2$ is impossible by $r \geq 4$; otherwise (4) gives $W \geq 4h - 2r \geq 2h - 2 > h + 1$.

With three occupied slices let their consecutive cardinalities be a, b, c , with b in the middle, and put $m = \max(a, c)$. The empty slice and (3) give

$$W \geq b + \max(a, h - b) + \max(c, h - b) + |a - c| \geq b + 2 \max(m, h - b) \geq h + m.$$

The middle inequality follows by splitting $h - b$ below, within, or above the interval with endpoints a, c . The final inequality follows by comparing b to $h - m$. If $m \geq 2$ the result follows. If $m = 1$, both endpoints are singletons, and replacing h by $h + 1$ in the displayed bound gives $W \geq h + 2$. $\square$

Lemma 8 (Long-cycle lifting). *If $h \geq 6$, H is h -regular, and $d(H) \geq h$, then for every $L \geq 5$, $d(H \square C_L) = h + 3$ and its only minimum selections are singletons.*

Proof. A singleton attains $h + 3$. Assume $W \leq h + 3$, so $r \leq h + 3$, and exclude every nonsingleton. If $k \geq 5$, (4) gives $W \geq 5h - 2r \geq 3h - 6 \geq h + 6$. If $k = 2$, (5) gives $W \geq 2h \geq h + 6$. If $k = 1$ and $r \geq 2$, then $W \geq h + 2r \geq h + 4$.

For $k = 3$, if the occupied projection has at most one edge, its weighted degree sum is at most $r - 1$, so $W \geq 3h - r + 1 \geq 2h - 2 \geq h + 4$. Otherwise the three slices form a path with sizes a, b, c . Its outward

empty neighbors are distinct. Set $u = a + c$ and $\sigma = 1_{a=1} + 1_{c=1}$. Keep the middle slice's selected sites and use (3) at the endpoints to get

$$W \geq 2h - b + u + \sigma \geq h - 3 + 2u + \sigma.$$

This is at least $h + 4$ for $u \geq 3$. If $u = 2$, both endpoints are singletons and the sharper bound gives

$$W \geq b + 2 \max(1, h + 1 - b) + 2 \geq h + 4.$$

For $k = 4$, the occupied projection is a union of paths. With at least two components, $\delta \geq 4$, so $W \geq 4h - 2r + 4 \geq h + 4$. With one component let a, d be its endpoint cardinalities. Then $\delta = a + d$. The empty contributions are at least $|a - d|$ when $L = 5$ and at least $a + d$ when $L \geq 6$. Consequently their sum with δ is at least $2 \max(a, d)$. If this is less than four, $a = d = 1$, and the two singleton bonuses in (4) supply the missing two. Again $W \geq 4h - 2r + 4 \geq h + 4$. $\square$

All-short tori

For $C_3^{\square a}$ the binary adjacency satisfies $A^2 = A$. Every x decomposes uniquely as $y + z$ with $y = Ax \in \text{im } A$ and $z = (A + I)x \in \ker A$. Its support union is $\text{supp } y \cup \text{supp } z$, hence

$$d(C_3^{\square a}) = \min\{d(\text{im } A), d(\ker A)\}.$$

The Hamming-graph code result of Fish, Key and Mwambene [5] gives $d(\text{im } A) = 2a$ and, for $a \geq 4$, $d(\ker A) = 2a + 1$. Their proof also records the small dual distances 3, 3, 6 for $a = 1, 2, 3$, respectively. In particular $d(C_3^3) = 6$ and $d(C_3^a) = 2a$ for $a \geq 4$. These are applications of existing results.

The remaining base $C_3^2 \square C_4$ has distance six by the following elementary argument. In $H = C_3^2$, singleton weight is five, pair weight is six, and every selection has weight at least its size. Assume $W \leq 5$ in $H \square C_4$. One occupied slice gives $w_H(x) + 2r \geq 7$; two give at least eight by (5), including the possible size pairs (3, 1), (3, 2), (4, 1). Four occupied slices have sizes at most two, and summing internal lower bounds four gives $W \geq 16 - 2r \geq 6$. With three slices all of size at most two, the proof of Lemma 7 applies with $h = 4$. The only other profiles are permutations of (3, 1, 1): the middle-three profile has occupied contribution at least $2 + 3 + 2 = 7$, while an endpoint-three profile has at least $3 + 1 + 4 = 8$. This excludes $W \leq 5$.

Starting from the one-dimensional bases C_3, C_4 of distance two, the base $C_3^2 \square C_4$, and the published all-triangle bases, repeated application of Lemma 7 proves $d \geq 2D$ for every all-short torus of dimension $D \geq 3$. To see equality, write its adjacency as $A = B + C$, where B sums triangle-coordinate adjacencies and C sums four-cycle adjacencies. Commutativity over $\mathbb{F}_2$ gives $A^2 = B$. For $x = Ae_v$, therefore, $Ax = Be_v$ is supported inside x , which has weight $2D$.

Analytic low-dimensional mixed bases

For $H \in \{C_3 \square C_3, C_3 \square C_4, C_4 \square C_4\}$ a singleton has weight five and a pair has weight at least six. Adjacent vertices have at most one common neighbor; nonadjacent vertices have at most two. This proves the pair bound from $|A(e_u + e_v)| = 8 - 2|N(u) \cap N(v)|$, adding the selected vertices when they are nonadjacent. Thus an analytic lower profile is

$$f(0), \dots, f(7) = (0, 5, 6, 3, 4, 5, 6, 7).$$

For a cyclic sequence of slice sizes define

$$B(r) = \sum_{r_t > 0} \max\{r_t, f(r_t) - r_{t-1} - r_{t+1}\} + \sum_{r_t = 0} |r_{t-1} - r_{t+1}|. \quad (6)$$

The slice inequality gives $W \geq B(r)$. We prove $B(r) \geq 8$ for every $L \geq 5$ and $2 \leq r = \sum r_t \leq 7$, without restricting L to a finite range.

With one occupied position, $B = f(r) + 2r \geq 9$. With two, each occupied position has a distinct private empty neighbor, so the empty contribution is at least r . Thus $B \geq 2r \geq 8$ if $r \geq 4$. If $r = 2$ or 3, both populations a, b are at most two, and their occupied contributions are at least $f(a) - b + f(b) - a = 8$.

With three occupied positions, a projection with no edges contributes at least nine internally. If there is one edge with populations a, b , its contribution is at least $g(a, b) = \max(a, f(a) - b) + \max(b, f(b) - a)$. For $a + b \geq 5$, $g \geq 5$ by selected cardinality. The remaining unordered pairs $(1, 1), (1, 2), (1, 3), (2, 2)$ give $g = 8, 8, 5, 8$, respectively. The isolated slice contributes at least three, proving $B \geq 8$. If there are two edges, the projection is a path with populations a, b, c . Its outward empty neighbors contribute $u = a + c$, hence

$$B \geq u + b + \max(a, f(a) - b) + \max(c, f(c) - b).$$

For $u \geq 4$, this is at least $2u + b \geq 9$. For $u = 2$, it is $b + 2 + 2\max(1, 5 - b) \geq 8$. For $u = 3$, it is $b + 3 + \max(1, 5 - b) + \max(2, 6 - b) \geq 10$.

With four occupied positions the degree deficit δ is at least two. The seven possible unordered cardinality partitions and their elementary lower bounds $\sum f(r_i) - 2r + 2$ are

$(r_i)_{i \in P}$	1111	2111	3111	2211	4111	3211	2221
lower bound	14	13	8	12	7	7	11.

This table merely substitutes seven integer partitions into the displayed inequality; it uses no enumeration program. For the two preliminary bounds seven, $\delta \geq 3$ suffices. If $\delta = 2$, the projection is one path with singleton endpoints. Up to reversal its profile is $(1, 4, 1, 1)$ or $(1, 3, 2, 1)$. Their occupied maxima are at least $(1, 4, 1, 4)$ and $(2, 3, 2, 3)$, respectively, both totaling ten.

Finally, with $k \geq 5$ occupied positions and $r \leq 7$, every size is at most three and at most one is three. Thus $\sum f(r_i) \geq 5k - 2$, and $B \geq 5k - 2 - 2r \geq 5k - 16 \geq 9$. All cases are covered. A nonsingleton selection of weight at most seven would have $2 \leq r \leq 7$ and therefore $W \geq 8$, a contradiction. Each of the three short-cross-section products with a long cycle consequently has distance seven and singleton-only minima.

Analytic two-dimensional strips and minimum classification

In fact, for every $a \geq 3$ and $L \geq 5$, $C_a \square C_L$ has distance five, with singleton-only minima except at $a = L = 5$. Slice along C_L and replace the preceding lower profile by $f(1) = 3$ and $f(s) = s$ for $s \geq 2$. Assume $W \leq 5$, so $r \leq 5$, and exclude nonsingletons.

One occupied slice gives $W \geq f(r) + 2r \geq 6$. With two occupied slices, distinct private empty neighbors contribute r , so $W \geq 2r \geq 6$ for $r \geq 3$. For $r = 2$, two nonadjacent singleton slices have unchanged internal weights summing to six; adjacent ones obey the exact inequality (5).

With three occupied slices and no occupied edges, the internal f -sum is at least seven. With one occupied edge, let its populations be a, b and the isolated population be c . If $a + b = 2$, the edge contributes at least four and $f(c) \geq 2$. If $a + b = 3$, the edge contributes at least three, which suffices when $c = 1$. When $c = 2$, the two empty gaps joining the components add at least $|a - c| + |b - c| \geq |a - b| = 1$, completing the lower bound six. A length-one gap gives the corresponding difference; a longer gap gives the boundary sum, at least the difference. If $a + b = 4$, necessarily $c = 1$, and the edge contributes at least four while the isolated slice contributes three. With two occupied edges, the projection is a path with endpoint population total u and middle population b . For $u \geq 3$, the selected and outward-empty contributions give $W \geq 2u + b \geq 7$. For $u = 2$, singleton endpoints give $W \geq b + 2 + 2\max(1, 3 - b) \geq 6$, for $b = 1, 2, 3$.

With four occupied slices and $r = 4$, the degree deficit gives $W \geq 12 - 2r + \delta \geq 6$. With $r = 5$, there is one double slice and three singleton slices. Let e be the number of occupied edges and d the induced degree of the double slice. Then $\delta = 10 - 2e - d$. Summing the raw slice bounds gives $11 - 2r + \delta$, and enforcing the double slice's two selected sites improves its raw bound $2 - d$ by exactly d . Hence $W \geq 11 - 2e \geq 7$ if $e \leq 2$. If $e = 3$, the occupied path is, up to reversal, $(2, 1, 1, 1)$ or $(1, 2, 1, 1)$. Their occupied maxima are $(2, 1, 1, 2)$ and $(1, 2, 1, 2)$, both summing to six.

Five occupied slices force five singletons. If their projection is proper, $W \geq 15 - 10 + \delta \geq 7$. Therefore only $L = 5$ with one selected vertex a_t in every slice can be a nonsingleton minimum. Since $W = r = 5$, each slice's binary Z output is supported only at its selected vertex. That output has even parity, as the internal cycle contributes two entries and the two neighboring singleton inputs also have even combined parity. It must therefore vanish:

$$A_{C_a} e_{a_t} = e_{a_{t-1}} + e_{a_{t+1}}.$$

The sequence (a_t) is a nonbacktracking closed length-five walk on C_a , which keeps a fixed orientation. Closure requires $a \mid 5$, hence $a = 5$. Its five offsets and two orientations yield precisely the ten diagonal selections. Each has zero odd neighborhood and weight five. This proves the distance and complete minimum classification for every $a \geq 3$, $L \geq 5$.

Dimension-three completion and induction

For $C_a \square C_L \square C_M$ with one short side and $L, M \geq 5$, assume $W \leq 7$. Slicing along either long side gives a cross-section of distance five by the strip result. With $r \leq 7$, the projection bounds for $k \geq 5$, $k = 4$, and $k \leq 3$ without adjacent empties are respectively at least 11, 8, 8. Thus both long directions have adjacent-empty pairs, and $W \geq r + 4$, so $r \leq 3$. Pairs have weight at least ten by the common-neighbor count in this degree-six graph. For a triple, slice along the short direction. One occupied slice contributes at least $5 + 6 = 11$. Two occupied slices have sizes one and two: on C_3 their occupied contribution is at least $10 - 3 = 7$ and the empty slice adds at least one; on C_4 , adjacent slices add three outward empty sites, and opposite slices already contribute at least ten. Three occupied singleton slices contribute at least nine on C_3 and eleven on C_4 . Only singleton selections remain. Together with the preceding all-long result, this proves the complete dimension-three base of Theorem 6.

Completion of Theorem 6. The all-short case was proved above. Induct on $D \geq 4$ in the case with a long side, choosing that side as the final factor. The cross-section has dimension $D - 1$, degree $h = 2D - 2 \geq 6$, and distance at least h , either by the all-short result or by induction. Lemma 8 gives distance $h + 3 = 2D + 1$ and singleton-only minima. Dimension three is supplied by the explicitly audited bases above. $\square$

Two-dimensional exception and quantum-code consequence

For $D = 2$ and at least one long side, the distance is five. The minimum multiplicity is N , except on $C_5 \square C_5$, where it is 35. The extra ten selections are $\{(t, c \pm t) : t \in \mathbb{Z}_5, c \in \mathbb{Z}_5\}$. The complete analytic proof above is also recorded in `higher_dimensions/dimension_two_analytic.md`. No minimum-multiplicity formula is asserted here for arbitrary-dimensional all-short mixtures.

For $G_D = C_3^{\square(D-1)} \square C_5$, Theorem 6 and the established CWS construction [2] yield a pure

$$[[5 \cdot 3^{D-1}, 1, 2D + 1]]$$

code for $D \geq 3$. Indeed the length- N repetition code has distance $N > 2D(2D + 1)$, the sufficient classical-distance condition, and $\text{span}\{|G_D\rangle, Z_{\text{all}}|G_D\rangle\}$ has stabilizer generated by $K_0 K_v$. A canonical K_v is a logical operator of weight $2D + 1$, showing exact distance. Independently, the 45-qubit instance has 392,176 distinct syndromes for all errors of weight at most three, proving purity and distance at least seven; K_v proves equality. The code rank and logical commutation checks are included in `code_application/verify_code.py`.

The 15-qubit instance is not new: the checked permutation $(a, b) \mapsto 5a + 3b \pmod{15}$ identifies its graph with the $[[15, 3, 5]]$ construction in Table I of Ref. [4]. The repetition word lies in that published classical word code, so our $[[15, 1, 5]]$ is a literal subcode. The CWS conversion itself is established, and no best-parameter, fixed-dimensional locality, or hardware advantage is claimed. The family has degree $2D$, distance logarithmic in its length, and rate $1/N$.

Research status and remaining priority questions

The equal-side two-dimensional distance-five family is expressly stated in Ref. [4], Example 6. Its Example 5 concerns quantum codes on open lattice fragments with distance $2D$, a different statement. Ref. [8] also covers rectangular two-dimensional lattices with both side lengths at least five and exhibits the diagonal minimum on the five-by-five torus. Existing classical adjacency-code results cover important all-short subfamilies. The inspected Cartesian section of Ref. [6] provides LCD preservation and codeword constructions; it does not supply the support-union lower bound in Lemma 8. The full 2017 cycle-product article [7] and the result behind the 2019 abstract remain unresolved priority checks. Accordingly, the present contribution is a concrete, reproducible proof candidate; publication novelty has not been established.

References

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